

ANALYSTLAB AFRICA

DATA ANALYTICS INTERNSHIP PROGRAM

Customer Behaviour & Business Performance Analysis

for ABC Communications Ltd

A Customer Churn Analysis Report

Prepared by

Adesola Sobambo

Dataset

Telco Customer Churn Dataset

Date

August 8, 2026

Table of Contents

Introduction	3
Part 1 – Business Understanding Report	4
Part 2 – Data Loading & Inspection.....	6
Part 4 – Business Insights	8
Part 5 – Business Recommendations	10
Conclusion	11

Note: Part 3 (detailed charts and technical analysis) is provided separately in the accompanying Excel workbook.

Introduction

This report presents a complete analysis of customer churn in a telecommunications company. The study examines a dataset of 7,043 customers to understand the current level of churn, identify the main factors associated with customers leaving, and provide practical recommendations for improving retention.

The report is structured as follows:

- **Part 1** – provides the business understanding of the churn problem.
- **Part 2** – covers the loading and inspection of the dataset.
- **Part 3** – detailed charts and technical analysis supporting this report, contained in the accompanying Excel workbook.
- **Part 4** – presents key business insights, risks, and opportunities derived from the analysis.
- **Part 5** – offers actionable recommendations based on the findings.

Part 1 – Business Understanding Report

1. What is Customer Churn?

Customer churn occurs when a customer stops doing business with a company. In the telecommunications industry, this usually means a subscriber cancels their mobile, internet, or bundled service and switches to a competitor or simply disconnects.

Churn is one of the most closely watched metrics in telecom because the cost of acquiring a new customer is typically much higher than the cost of retaining an existing one. High churn directly reduces recurring revenue and increases marketing and onboarding expenses.

2. Business Context and Problem Statement

This project examines a telecommunications company's customer base of **7,043** subscribers. Of these, **1,869** customers (approximately **26.5%**) have already churned.

The core business problem: *Why are customers leaving, and which customers are most at risk of leaving in the future?*

Understanding the answer will allow the company to design more effective retention strategies, protect revenue, and improve long-term customer lifetime value.

3. Why This Matters to the Business

- Telecom services generate ongoing monthly revenue. Every lost customer creates an immediate and recurring revenue gap.
- Competition in the market is intense; customers can switch providers relatively easily, especially those on flexible contracts.
- Early identification of at-risk customers gives the business a chance to intervene before the customer decides to leave.
- Insights from churn analysis can also guide product packaging, pricing, and customer-service improvements.

4. Project Objectives

1. Establish a clear picture of the current churn situation.
2. Explore customer and service characteristics recorded in the data.
3. Identify patterns associated with higher or lower likelihood of churn.
4. Translate findings into practical business understanding that can support retention decisions.

5. Data Available for Analysis

The dataset contains one row per customer and includes information on:

- Demographics (gender, senior citizen status, partner, dependents)
- Account details (tenure, contract type, payment method, paperless billing)
- Services subscribed (phone, internet type, online security, tech support, streaming, etc.)
- Charges (monthly charges and total charges)
- The target variable: whether the customer churned (Yes/No)

This information provides a solid foundation for understanding customer behaviour and service usage patterns related to churn.

Part 2 – Data Loading & Inspection

1. Dataset Size

Item	Value
Number of rows (customers)	7,043
Number of columns	21

2. Columns and Data Types

Column	Data Type	Description
CustomerID	Text	Unique customer identifier
Gender	Text	Male / Female
SeniorCitizen	Integer	0 = No, 1 = Yes
Partner	Text	Yes / No
Dependents	Text	Yes / No
Tenure	Integer	Number of months with the company
PhoneService	Text	Yes / No
MultipleLines	Text	Yes / No / No phone service
InternetService	Text	DSL / Fiber optic / No
OnlineSecurity	Text	Yes / No / No internet service
OnlineBackup	Text	Yes / No / No internet service
DeviceProtection	Text	Yes / No / No internet service
TechSupport	Text	Yes / No / No internet service
StreamingTV	Text	Yes / No / No internet service
StreamingMovies	Text	Yes / No / No internet service
Contract	Text	Month-to-month / One year / Two year
PaperlessBilling	Text	Yes / No
PaymentMethod	Text	Electronic check, Mailed check, Bank transfer, Credit card
MonthlyCharges	Float	Current monthly charge
TotalCharges	Float	Total amount charged to date
Churn	Text	Yes / No (target variable)

3. Missing Values

After cleaning, the dataset has **0 missing values**.

Note on cleaning performed: TotalCharges originally contained 11 blank values (all belonging to customers with Tenure = 0). These were replaced with 0, which is logical because new customers have not yet been billed. The column was then converted to numeric (float).

4. Duplicates

Check	Result
Duplicate rows	0
Duplicate CustomerIDs	0

The dataset contains unique customer records only.

5. Summary Statistics (Numeric Columns)

Statistic	SeniorCitizen	Tenure	MonthlyCharges	TotalCharges
Count	7,043	7,043	7,043	7,043
Mean	0.16	32.37	64.76	2,279.73
Std Dev	0.37	24.56	30.09	2,266.79
Minimum	0	0	18.25	0.00
25th Percentile	0	9	35.50	398.55
Median (50%)	0	29	70.35	1,394.55
75th Percentile	0	55	89.85	3,786.60
Maximum	1	72	118.75	8,684.80

Quick Observations

- About 16% of customers are senior citizens.
- Average customer tenure is roughly 32 months (just over 2.5 years).
- Monthly charges range from ₦18.25 to ₦118.75.
- TotalCharges starts at 0 (new customers) and goes up to ₦8,685.

6. Categorical Variables – Unique Value Counts

Most categorical columns have 2 or 3 clean levels. PaymentMethod has 4 categories. CustomerID is unique for every row (7,043 distinct values). No unexpected or messy category labels were found.

Part 4 – Business Insights

Five Key Insights

1. **Contract type is the strongest driver of churn.**

Customers on month-to-month contracts churn at 42.7%, compared to 11.3% for one-year contracts and only 2.8% for two-year contracts. Longer contracts are clearly linked to much better retention.

2. **Fiber optic customers have a very high churn rate.**

Fiber optic users churn at 41.9%, more than double the rate of DSL customers (19.0%) and far higher than customers with no internet service (7.4%). This is a major risk area despite fiber being a premium service.

3. **Payment method has a strong impact on churn.**

Customers who pay by electronic check churn at 45.3%, while those using automatic payments (bank transfer or credit card) churn at only 15–17%. Manual payment methods are associated with significantly higher dropout.

4. **New customers are the most likely to leave.**

Almost half of customers (47.4%) churn within their first 12 months. Churn falls steadily as tenure increases, reaching just 9.5% for customers who stay 49–72 months. The first year is the critical risk window.

5. **Higher monthly charges are linked to higher churn.**

Churned customers have a higher average monthly charge (N74.4) than retained customers (N61.3). The box plot also shows that customers who leave tend to have higher median spending.

Three Business Risks

1. **Heavy reliance on month-to-month contracts.**

A large share of the customer base is on the contract type with the highest churn rate, creating ongoing revenue instability.

2. **High churn in the fiber optic segment.**

Fiber optic is likely a high-revenue service, but the 41.9% churn rate threatens its profitability and may indicate pricing or service problems.

3. **Weak retention of new customers.**

Losing nearly 50% of customers in the first year means high acquisition costs are not being converted into long-term value.

Three Business Opportunities

1. **Convert month-to-month customers to longer contracts.**

Moving even a portion of month-to-month customers onto one- or two-year contracts could significantly reduce overall churn.

2. **Promote automatic payment methods.**

Encouraging customers to switch from electronic check to automatic payments (bank transfer or credit card) can improve retention at relatively low cost.

Evidence: *Electronic check customers churn at 45.3%, while automatic payment customers churn at only 15–17%.*

3. **Strengthen the first-year customer experience.**

A focused onboarding and early-engagement program for new customers has strong potential to reduce the high volume of early churn.

Part 5 – Business Recommendations

1. Migrate month-to-month customers to longer contracts

Offer incentives (discounts, free months, or add-ons) to encourage customers on month-to-month plans to move to one-year or two-year contracts.

Evidence: Month-to-month customers churn at 42.7%, compared with 11.3% (one-year) and 2.8% (two-year).

2. Address high churn in the Fiber Optic segment

Investigate pricing, service quality, and competitor offers for fiber optic customers, and design a targeted retention program for this group.

Evidence: Fiber optic customers have a 41.9% churn rate, more than double that of DSL customers.

3. Encourage customers to switch to automatic payments

Run a campaign to move customers from electronic check to automatic payment methods (bank transfer or credit card), possibly with a small incentive.

Evidence: Electronic check users churn at 45.3%, while automatic payment users churn at only 15–17%.

4. Focus retention efforts on the first 12 months

Introduce a structured onboarding and early-engagement program for new customers during their first year.

Evidence: 47.4% of customers churn within the first 12 months — by far the highest risk period.

5. Review pricing and value for higher-paying customers

Examine whether higher monthly charges are delivering enough perceived value, and consider loyalty offers or package reviews for customers in the higher charge bands.

Evidence: Churned customers have higher average monthly charges (N74.4) than retained customers (N61.3), and the box plot shows higher median spend among those who left.

Conclusion

The analysis shows that 26.5% of customers have churned. The strongest drivers of churn are month-to-month contracts, fiber optic service, electronic check payments, and short customer tenure (especially the first 12 months). Higher monthly charges are also associated with a greater likelihood of leaving.

These findings highlight clear areas of risk, particularly the company's reliance on flexible contracts and the high dropout rate among new customers. At the same time, they point to practical opportunities: converting customers to longer contracts, promoting automatic payments, and improving the first-year experience.

By focusing retention efforts on these priority areas, the company can reduce churn, protect recurring revenue, and improve long-term customer value.